

Education

- 2015-2020 **Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering**
Indian Institute of Technology Madras, Chennai, India CGPA: 8.78
- 2015 **XII - Karnataka Board, KLE Society's Independent PU College, Bangalore** 97.30 %
- 2013 **X - ICSE, B P Indian Public School, Bangalore** 96.33%

Professional Experience

- Software Engineer, Level IV, Google LLC, New York**
- Dec 2022 - Present **Software Engineer, Level IV, Google India Pvt Ltd, Bangalore**
- Aug 2020 - Nov 2022 { Developing intelligent features for the Google Workspace Editors (Docs, Slides, Keep, etc) using my expertise on the products' client-side software, supporting tools and libraries, and natural language processing infrastructure.
{ Using cutting-edge frontend tools like Web Assembly and Emscripten, and Google-internal technologies like j2Cl, client-side cross-platform frameworks and build systems, to develop user-facing features such as spellcheck in encrypted documents for five languages and writing style suggestions for English text.
{ Formulating technical designs for independent end-to-end problems, driving cross-team collaboration, upholding software reliability practices, technical-debt resolution and documentation, and proactively identifying areas of future work.
{ Guiding junior engineers on programming and software design tasks to enable timely delivery of products to customers.
- May 2019 - July 2019 **Software Engineering Intern, Google India Pvt Ltd, Bangalore**
- Worked on the Editors client-side software infrastructure to develop a user interface with control options to undo or provide feedback on the correction and a logging framework, for the Google Docs text auto-correction feature.
- May 2018 - July 2018 **Research Intern, Big Data Experience Labs, Adobe Research, Bangalore**
- Developed a mobile application for Text to Scene Conversion in Augmented Reality, based on novel research techniques for prediction of three-dimensional object sizes and positions from textual features.

Research Experience

- Sep 2019 - May 2020 **Paraphrase Generation with a Bilingual Model and Continuous Embeddings** *Master's Thesis, Language Technologies Institute, Carnegie Mellon University*
- Machinated a novel technique for paraphrase generation using the [von Mises-Fisher \(vMF\) Loss](#) on a transformer network, and showed that it produces superior paraphrases as compared to the log-likelihood model by employing bilingual data to induce zero-shot paraphrasing, guided by [Prof. Yulia Tsvetkov](#).
- May 2017 - July 2017 **Cognitive Approach to Natural Language Processing**
- Research Intern, Department of Computer Science and Automation, Indian Institute of Science (IISc), Bangalore*
- Developed a cognitive text parser that combines syntactic and semantic approaches, to process textual data into cognitive structural representations, to be used as a feature extractor for downstream NLP tasks, and demonstrated the correlation of the extracted cognitive features with semantic and syntactic text features, guided by [Prof. Veni Madhavan](#).

Publications and Patents

- [Publication and Poster] **Improving the Diversity of Unsupervised Paraphrasing with Embedding Outputs (Paper, Poster)**
Monisha Jegadeesan, Sachin Kumar, John Wieting, Yulia Tsvetkov
In [Workshop on Multilingual Representation Learning](#),
The 2021 Conference on Empirical Methods in Natural Language Processing ([EMNLP 2021](#))
- [Publication and Poster] **Adversarial Demotion of Gender Bias in Natural Language Generation (Paper, Poster)**
Monisha Jegadeesan
In [ACM CODS-COMAD 2020](#) - Young Researchers' Symposium
- [Poster] **ARComposer: Authoring Augmented Reality Experiences through Text (Poster)**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
In ACM User Interface Software and Technology Symposium 2019 ([ACM UIST 2019](#))
- [Filed Patent] **Visualizing Natural Language through 3D Scenes in Augmented Reality**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
Filed at the US PTO (Application Number: 16/247,235)

Projects

Real-Time AI-Powered Customer Support Chatbot (2024)

Designed and deployed a scalable AI chatbot leveraging LLM-based systems and retrieval-augmented generation (RAG) to deliver instant responses to customer queries using the company knowledge base. Built vector embeddings and managed feature extraction from unstructured data to enable real-time inference. Integrated the solution into production, resulting in a 35% reduction in human agent workload and significantly improved response latency, demonstrating end-to-end ML lifecycle ownership from problem framing to deployment and continuous improvement.

Tech: Python, LangChain, OpenAI API, ChromaDB, LLM Integration

Predictive Maintenance Model for Industrial Machinery (2024)

Engineered and deployed a predictive maintenance pipeline for manufacturing equipment by processing large-scale IoT sensor data including temperature, vibration, and pressure. Extracted time-series features and trained a Random Forest model to forecast machine failures, then integrated it into a production environment with monitoring to track model performance and detect data drift. The model reduced unexpected downtime by 28% and delivered measurable operational cost savings, showcasing pragmatic ML application and full lifecycle management from modeling through deployment and iteration.

Tech: Python, Random Forest, Time-Series Feature Extraction, Sensor Data Processing

Customer Sentiment Classification System for E-commerce (2024)

Developed and deployed an NLP-based sentiment analysis model to classify customer sentiment from large-scale, scraped e-commerce product reviews. Executed feature engineering through TF-IDF vectorization and implemented a Logistic Regression classifier optimized for accuracy and production reliability. Continuously monitored model outputs for drift and iterated based on feedback to maintain an 89% classification accuracy, contributing directly to product quality improvement through actionable insights.

Tech: Python, NLTK, TF-IDF, Logistic Regression, NLP

Sales Forecasting and Inventory Optimization Model (2024)

Built and deployed scalable time-series forecasting models including ARIMA and Facebook Prophet to predict monthly retail sales across multiple product categories. Translated historical sales data into meaningful features and implemented pipelines for batch inference in production. Reduced forecasting error by 25%, enabling optimized inventory planning and reduced stockouts. Emphasized model reliability, latency considerations, and continuous monitoring to ensure real-world impact and iterative improvements in operational performance.

Tech: Python, Prophet, ARIMA, Pandas, Time-Series Forecasting

About :

I am an AI / Machine Learning Engineer specializing in designing, building, deploying, and improving scalable machine learning models that solve real-world problems and deliver measurable business impact. Proficient in Python and ML frameworks such as TensorFlow, PyTorch, and scikit-learn, with strong engineering fundamentals and experience in end-to-end ML lifecycle including data preprocessing, feature engineering, model optimization, deployment, and monitoring. Experienced in deploying reliable, scalable models into production environments using modern tooling and best practices including Docker, CI/CD, and cloud infrastructure. Skilled in working with large-scale data, reasoning about trade-offs, and maintaining model performance, data quality, and drift monitoring. Passionate about owning the full ML lifecycle, collaborating with cross-functional teams, and focusing on practical, production-ready solutions that optimize operations and product performance.

